1 Objective and Introduction

- **Objective**
  Understand the definition of Lie algebra representations, focusing specifically on $\mathfrak{sl}_2$. Explore the properties of these representations through a combinatorial lens.

- **Representations of Lie Algebras**
  A Lie algebra $\mathfrak{g}$ is a vector space with a Lie bracket

  $$[-,-] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g},$$

  which is bilinear, alternating, and satisfies the Jacobi identity. Given a vector space V, a representation of a Lie algebra is a map

  $$\rho : \mathfrak{g} \to \mathrm{End}(V),$$

  where the bracket in $\mathrm{End}(V)$ is the matrix commutator and the map preserves the Lie bracket structure.

- **The Lie Algebra $\mathfrak{sl}_2$**
  The Lie algebra $\mathfrak{sl}_2(\mathbb{C})$ is

  $$\mathfrak{sl}_2 = \{\, A \in M_2(\mathbb{C}) : \ \mathrm{tr}(A) = 0 \,\}$$

  It is generated by:

  $$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

  With the Lie bracket $[X, Y] = XY - YX$, the generators satisfy the relations:

  $$[E, F] = H, \quad [H, E] = 2E, \quad [H, F] = -2F.$$

- So we may outline how we're going to complete our objective:

  (a) Understand the irreducible representations of $\mathfrak{sl}_2$ and view examples.

  (b) Visualize the representations as graphs and connect them with crystal base theory.

  (c) Analyze tensor products of representations, derive the Clebsch-Gordan rule.

  (d) Connect the representation theory of $\mathfrak{sl}_2$ to crystals of $\{\,1, 2\,\}$-words.

2 Classifying Irreducible Representations

Theorem 1. *If V is an irreducible representation of $\mathfrak{sl}_2$, it decomposes into weight spaces as*:

$$V = \bigoplus_{\alpha \in \operatorname{Spec} H} V_\alpha,$$

where V_α are the 1-dimensional eigenspaces of H.

Each irreducible representation can be visualized as a graph whose vertices correspond to the eigenspaces of H and the directed edges to the actions of the generators.

Example 1. The *trivial representation* is $\mathbb{C}$ where E, F, and H map to "multiplication by 0". The only eigenvalue of H is 0, so the decomposition is trivially $V = V_0$.

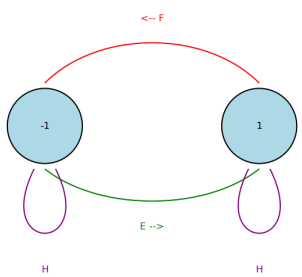


Figure 1: Graph of the standard representation $V^{(1)}$

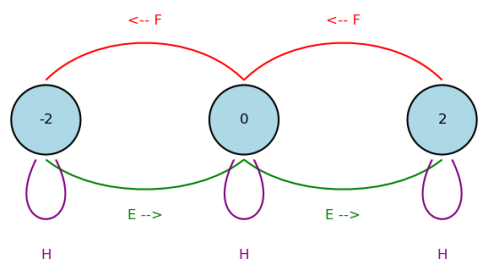


Figure 2: Graph of the adjoint representation $V^{(2)}$

Example 2. For the *standard representation*, the map is

$$\rho : \mathfrak{sl}_2 \to \operatorname{End}(\mathbb{C}^2)$$

which happens to be the inclusion, i.e. the matrices map to themselves. In this case,

$$He_1 = e_1 \quad \text{and} \quad He_2 = -e_2,$$

resulting in the decomposition $V^{(1)} = \mathbb{C}^2 = \operatorname{gen}(e_1) \oplus \operatorname{gen}(e_2) = V_1 \oplus V_{-1}$.

Example 3. The *adjoint representation* is determined by the adjoint map

$$\operatorname{Ad} : \mathfrak{sl}_2 \to \operatorname{End}(\mathfrak{sl}_2), \quad \operatorname{Ad}_X(Y) = [X, Y].$$

Observe that via the commutation relations

$$[E, F] = H, \quad [H, E] = 2E, \quad [H, F] = -2F,$$

we have that $V^{(2)} = V_2 \oplus V_0 \oplus V_{-2}$.

3 Tensor Products

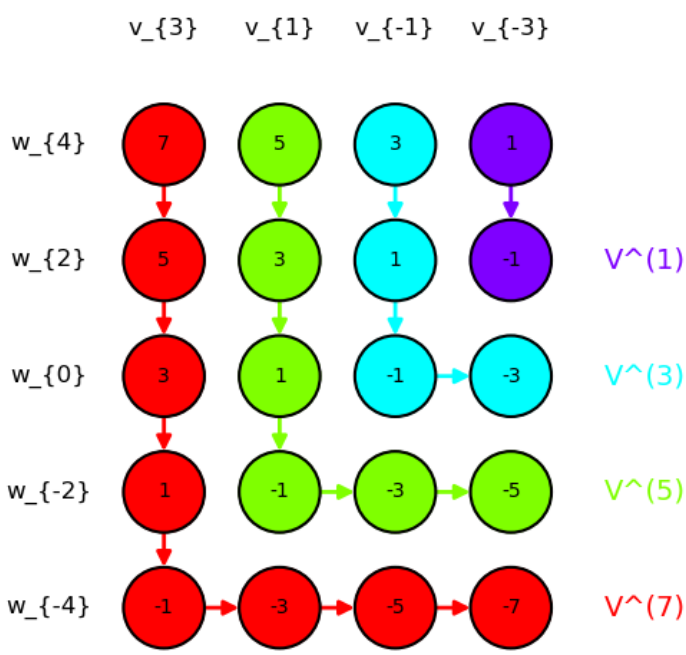
We may tensor representations to obtain new representations. The Clebsch-Gordan rule gives us a way to decompose the tensor product $V^{(n)} \otimes V^{(m)}$ into irreducibles.

Theorem 2. *If $n \geqslant m$, we have*:

$$V^{(n)} \otimes V^{(m)} \cong V^{(n+m)} \oplus V^{(n+m-2)} \oplus V^{(n+m-4)} \oplus \cdots \oplus V^{(n-m)}$$

Graphically, this operation can be done by arranging $V^{(n)}$ and $V^{(m)}$ in a grid and concatenating the graphs together.

Example 4. If we take the tensor product of $V^{(4)} \otimes V^{(3)}$, we can arrange the basis vectors into a grid and concatanate as follows:



Here we only draw the arrows representing the action of F (since E arrows point in the opposite direction, and the H eigenvalues can be read directly). Each distinct path corresponds to one irreducible component, verifying the Clebsch-Gordan decomposition:

$$V^{(4)} \otimes V^{(3)} \cong V^{(7)} \oplus V^{(5)} \oplus V^{(3)} \oplus V^{(1)}.$$

Exercise 1. Try your hand at computing $(V^{(1)})^{\otimes 3}$.

4 Connection with $\{1, 2\}$-words

A word in the alphabet $\{1, 2\}$ is a string comprised of only those two characters.

Example 5. The strings 12221112121, 22221112121, and 22212112211 are valid words!

We may connect these words to the standard representation $V^{(1)}$ of $\mathfrak{sl}_2$ via the correspondence for the basis vectors $1 \leftrightarrow v_1, \quad 2 \leftrightarrow v_{-1}$.

Thus, a word of length n corresponds directly to a basis vector in the n-fold tensor product $(V^{(1)})^{\otimes n}$.

The lowering operator F acts on these words as follows:

(a) Replace all 1's with right parentheses) and all 2's with left parentheses (.

(b) Greedily match any adjacent parentheses pairs ().

(c) The F operator finds the **rightmost unmatched** 1 (which corresponds to an unmatched)) and turns it into a 2 (which is an (). If there is no unmatched 1, F annihilates the word.

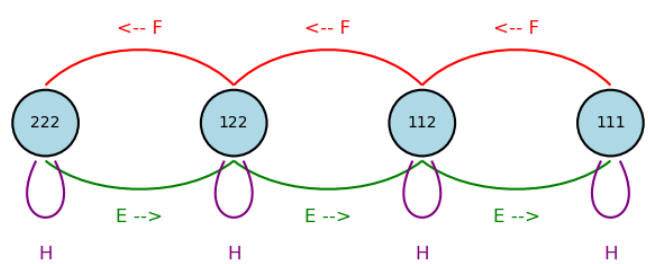
The raising operator E acts in the opposite way.

Exercise 2. Apply the F operator to the words 211, 1122, and 12221112121.

The complete connection with words is established by the following theorem:

Theorem 3. *The irreducible representation $V^{(n)}$ is isomorphic to the nth symmetric power of the standard representation,* $\mathrm{Sym}^n(V^{(1)})$.

For example, this provides a clear correspondence between the weight diagram for $V^{(3)}$ and the crystal graph for words of length 3:



Exercise 3. Try your hand at drawing the word graphs corresponding to the decomposition into irreducibles of $(V^{(1)})^{\otimes 3}$.